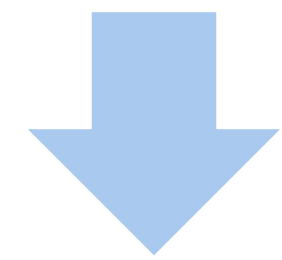
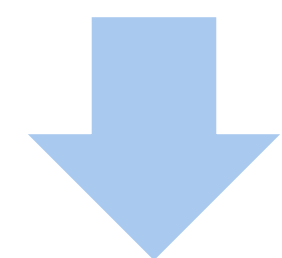


## BUILD

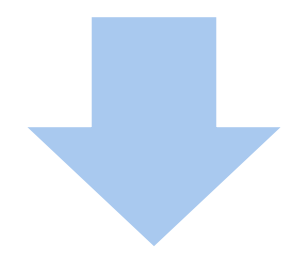
**Discharge notes**  
328,839 MIMIC-IV



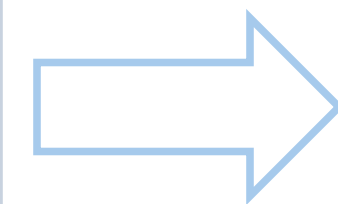
**Qwen extraction**  
cause → effect triples



**UMLS normalization**  
merge synonyms to CUIs



**Candidate causal KG**  
4.3K concepts / 19K edges



## AUDIT

### Which validation signals deserve trust?

Scored against SemMedDB literature causal predicates as an independent reference.

**WORKS**

#### Co-occurrence lift

EHR support signal  
filters weakly supported co-mentions

**FAILS**

#### Coding-order time

Weak direction signal  
54.2% vs 73.6%; coding order is not onset

**PARTIAL**

#### Literature evidence

Independent reference  
74% dir. agree; ~30% coverage

**Takeaway: lift supports candidates;  
coding-order time should not orient edges.**



## APPLY

**EHR-supported  
causal-candidate KG**



**Support-aware causal  
RAG + demotion**



**Held-out patient  
causal question**

**+9.9 pts  
vs unvalidated graph**